

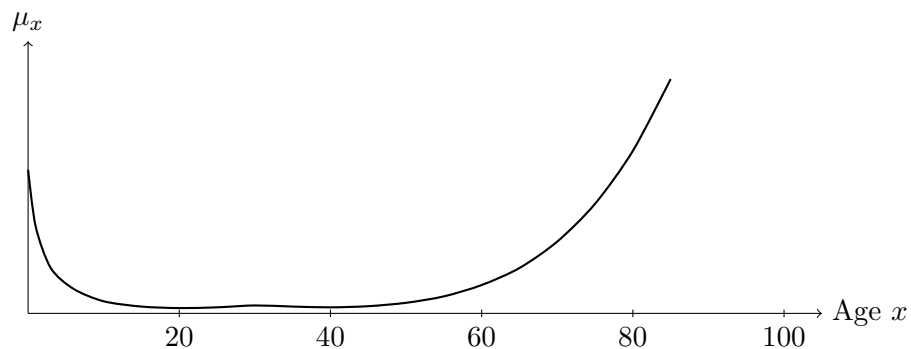
Exercise 3.1

Sketch the following as functions of age x for a typical human population, and comment on the major features:

- (a) μ_x ,
- (b) l_x , and
- (c) d_x .

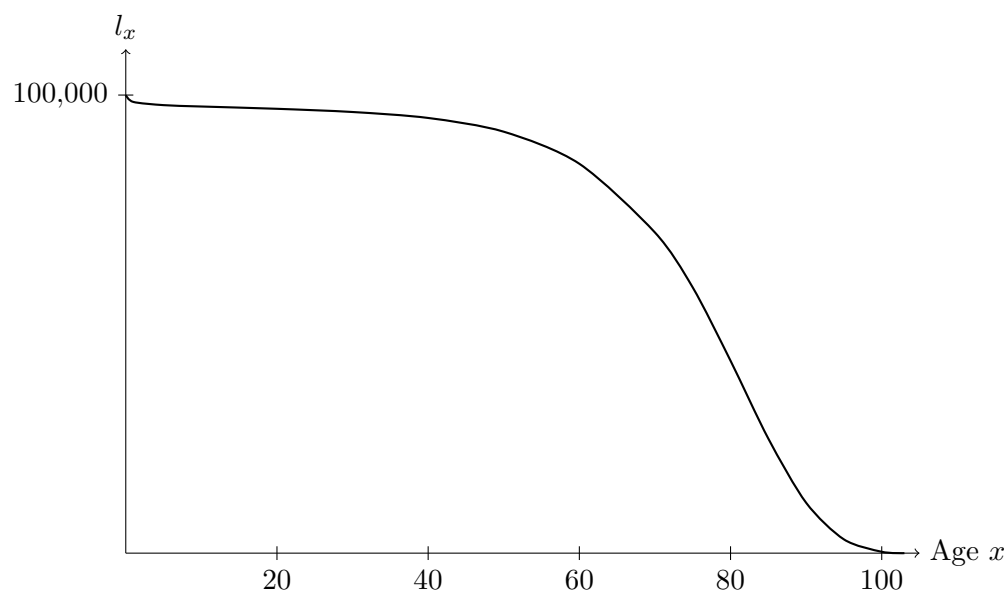
Solution

- (a) Force of mortality μ_x



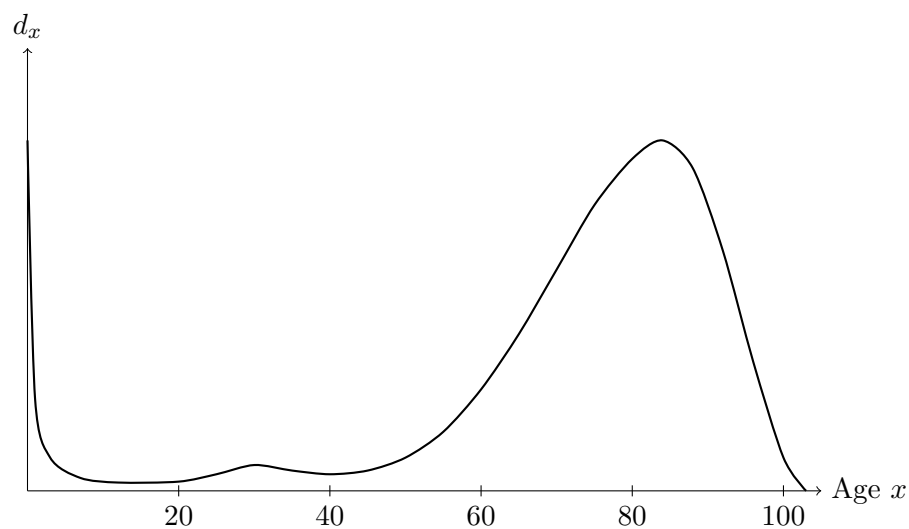
The curve is approximately U-shaped. Mortality is relatively high immediately after birth, but it decreases rapidly throughout infancy and childhood. It is very low at young ages, sometimes with a small accident-related hump in early adulthood. From middle age onward, mortality increases rapidly and is often approximated by Gompertz's law, under which the force of mortality grows exponentially with age.

(b) Number of survivors l_x



The function l_x is nonincreasing because it represents the number of members of the original cohort who survive to exact age x . It begins at the chosen radix, commonly $l_0 = 100,000$. The decline is usually small through childhood and early adulthood, becomes faster at older ages, and approaches zero at the limiting ages of the population.

(c) Number of deaths d_x



For a life table,

$$d_x = l_x - l_{x+1} = l_x q_x.$$

There may be a noticeable concentration of deaths near birth, followed by very few deaths during childhood. The number of deaths then rises at adult and old ages, reaching a modal age at death before declining to zero as the remaining cohort is exhausted. Thus, even though q_x generally increases at advanced ages, d_x eventually decreases because very few lives remain exposed to the risk of death.